

# Oluwatobi Melvyn Mayungbo

AI Engineer, LLM and Retrieval Systems

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## SUMMARY

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AI engineer with a B.Sc. in Statistics who builds tools for agents and makes them reliable. Built an MCP server with six read-only tools and a resilient HTTP layer with caching, rate limits and retries, a retrieval assistant with refusal guardrails and a separate evaluation harness, and a React review screen for a deployed reconciliation system.

## SKILLS

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**AI and LLM Systems:** Model Context Protocol (MCP), LLM Evaluation and Guardrails, Retrieval-Augmented Generation (RAG), Prompt Engineering, Hybrid Search (BM25, Dense Embeddings, RRF), Citation Verification, Hugging Face, ONNX Runtime

**Engineering and MLOps:** Python, FastAPI, React, Docker, Pytest, CI/CD (GitHub Actions), SQL, DuckDB, PostgreSQL, Redis, GCP Cloud Run, AWS ECS Fargate, Railway, Prefect, dbt-core, MLflow, Evidently, Git, Streamlit

**Machine Learning:** PyTorch, Scikit-learn, LightGBM, XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers), Model Calibration (Isotonic Regression, Temperature Scaling), SHAP, Feature Engineering

**Statistics and Causal Inference:** Statistical Inference, Experiment Design, Power Analysis, CUPED, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, Time-Series Backtesting, Logistic Regression, WOE Scorecards

**Payments and Risk:** Transaction Fraud Detection, Credit Risk Decisioning, Payment Reconciliation, Double-Entry Ledgers and Idempotency, Regulatory Compliance (CBN/NDPA), Demand Forecasting, Web3 Fraud Risk Scoring

## PROJECTS

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### web3-risk-mcp: Web3 Fraud Risk MCP Server | [GitHub](#)

*Tech: Python, Model Context Protocol (MCP), Etherscan V2, GoPlus, DexScreener, Docker, GitHub Actions*

- Built an MCP server that gives AI assistants (Claude Desktop, Cursor) six read-only tools to check a crypto wallet, token or smart contract for risk before anyone interacts with it, returning a 0 to 100 risk score with a reason for every point, across Ethereum, Base, Arbitrum One, Polygon PoS and BNB Chain.
- Built a resilient HTTP layer with caching, per-source rate limits and retries with exponential backoff and jitter; one failing source never breaks an investigation, and missing data lowers confidence rather than the score.
- Assembled an evaluation set of 34 hand-checked addresses (12 risky, 22 safe) with a record and replay harness reporting ROC AUC, precision and recall.

### Fintech Compliance RAG Assistant | [GitHub](#) | [Live App](#)

*Tech: Python, Streamlit, BM25, ONNX Runtime, Reciprocal Rank Fusion, FastAPI, Docker, GCP Cloud Run*

- Designed generation to fail safely under uncertainty: a sentinel-token refusal mechanism enforced across the prompt, output guardrail and evaluation harness, plus post-generation citation verification that strips any marker the model cited but never retrieved.
- Ran retrieval and generation evaluation as separate metrics to localise failure by component; traced a 0.40 refusal-correctness result to a lexical-overlap limitation in the offline stub judge and flagged re-validation against a live provider as the next step instead of overstating readiness.
- Deployed on GCP Cloud Run via Docker within a ~225MB peak memory footprint (2GB budget), with per-session and daily request caps in application code to bound API cost on a public endpoint.

### Reckon: Payment Reconciliation and Review System | [GitHub](#) | [Live App](#)

*Tech: Python, PyTorch, DuckDB, FastAPI, React, Docker, Railway, PostgreSQL*

- Built and deployed a payment reconciliation system processing multi-source transaction feeds to detect uncollected revenue, duplicate settlements and payout discrepancies, combining deterministic matching rules with ML-assisted discrepancy resolution.
- Designed a human-in-the-loop review interface in React showing match confidence scores, audit trails and suggested resolution actions to streamline exception handling.

## EXPERIENCE

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**Machine Learning Instructor, SQI College of ICT**

March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
- Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.

**Internal Auditor, NISER**

May 2024 to July 2024

- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

**EDUCATION**

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**B.Sc. Statistics, University of Ibadan**

January 2021 to March 2025

**Professional Diploma in Artificial Intelligence, SQI College of ICT**

July 2025 to Present

**CERTIFICATIONS**

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- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action